

PRANOSH M

Aspiring ML Engineer & Full-Stack Developer

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PROFESSIONAL SUMMARY

Third-year Computer Science Engineering student at Panimalar Engineering College with hands-on experience building end-to-end intelligent systems — spanning real-time computer vision (YOLOv8, optical flow), multi-agent LLM orchestration, and cryptographic backend infrastructure. Backend AI Engineering intern at FlyRank AI. My goal is to engineer scalable, production-ready systems where full-stack depth and ML capability meet.

TECHNICAL SKILLS

Programming Languages:	Python, Java, JavaScript, C
ML & AI:	YOLOv8, DeepSORT, OpenCV, Hugging Face Transformers, LangChain, LLM Integration, Multi-Agent Architecture, PyTorch (familiar), scikit-learn (familiar)
Frontend:	React, HTML, CSS, Tailwind CSS
Backend:	Node.js, Express.js, FastAPI, REST APIs
Databases:	SQLite, SQL, MongoDB
DevOps & Tools:	Git, GitHub, GitLab

INTERNSHIP EXPERIENCE

Backend AI Engineer Intern

Remote | Jul – Aug 2026 (6 weeks)

FlyRank AI

- Authored API contracts for AI-powered backend services — specifying endpoints, payload schemas, status codes, and error responses across 4 workflow stages before a single line of implementation was written.
- Converted ambiguous AI feature briefs into fully scoped backend tasks with defined inputs, outputs, constraints, and failure cases — covering structured-output pipelines and multi-step agent loops.
- Constructed retrieval-backed answer flows with source grounding; assessed output quality against a 3-criteria rubric (accuracy, relevance, groundedness) and flagged unsupported responses for remediation.
- Produced evaluation sets, limitation reports, and production monitoring documentation — covering credential isolation, retry strategies, rate limiting, and observability for 2 deployed AI workflows.

PROJECTS

CrowdAlert — AI Crowd Monitoring & Emergency Alert System

Python · FastAPI · React · YOLOv8 · DeepSORT · OpenCV · CUDA · Twilio · SQLite · ReportLab

- Deployed YOLOv8m with CLAHE low-light preprocessing and CUDA-accelerated inference, achieving real-time person detection across 3 input sources (webcam, video file, CCTV stream); SmartFrameProcessor adaptive frame-skipping maintained pipeline throughput under high CPU load without dropping the detection loop.
- Constructed a 6-class crowd behaviour classifier (normal, loitering, gathering, panic, stampede, fighting) using Farneback dense optical flow — classifying by motion dynamics rather than headcount, eliminating false alarms in dense-but-calm environments.
- Built a severity-aware alert engine dispatching sub-second simultaneous alerts across 3 channels (SMS, voice call, email) with cooldown management to prevent alert spam; tracked full incident lifecycle — detection to exportable PDF/CSV report — with SQLite-backed history.

Multi-Agent Research Intelligence Platform

Python · React · REST API · LLMs · LangChain · Multi-Agent Architecture · Semantic Scholar API · arXiv API · OpenAlex API

- Architected a 4-agent system (Discovery, Analysis, Citation, Summarisation) with a central orchestration layer that dynamically determines agent execution order based on query context — reducing manual literature review workflow to a single prompt.
- Built a Research Discovery Agent aggregating 3 academic APIs (Semantic Scholar, arXiv, OpenAlex) with relevance-ranked pagination and per-paper metadata extraction — cutting multi-source literature search from hours of

manual querying to a single prompt.

- Engineered a Citation Generator supporting 4 formats (APA, MLA, IEEE, Chicago) with bulk export to .txt and .bib — eliminating manual citation formatting across the full academic writing workflow.

Tamper-Proof Welfare Disbursement Validation System

Node.js · Express.js · React · REST API · SHA-256 Cryptography · SQLite

- Designed a 3-gate cryptographic verification pipeline enforcing strict sequential execution — no gate skippable or bypassable via UI — reducing disbursement fraud surface to zero at the architecture level.
- Built an immutable SHA-256 hash-linked ledger where every record stores its predecessor's hash — making any post-hoc tampering mathematically detectable in O(n) audit time, with zero blockchain infrastructure required.
- Implemented an automatic system freeze on ledger mismatch or budget exhaustion, with a real-time admin dashboard and one-click audit CSV export for compliance reporting.

EDUCATION

B.E — Computer Science Engineering

Panimalar Engineering College, Chennai | 2024 – 2028 | Currently 3rd Year | CGPA: 7.95 / 10

- Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Linear Algebra

Full-Stack Development with GenAI Honours Program

NIIT | Expected Sep 2026

Higher Secondary Certificate (HSC)

Velammal Vidhyalaya Alapakkam, Chennai | 2024 | Score: 82.6%

CERTIFICATIONS & ADDITIONAL INFORMATION

Cloud Computing Internship — EI Systems (Robokwik Partner)

Languages: English (Professional) | Tamil (Native)